

Word Embeddings and text Classification

Problem Statement:

Develop an approach to effectively train a feedforward neural network for text classification while simultaneously deriving meaningful word embeddings. The objective is to build a neural network model that not only classifies text accurately but also learns high-quality word representations, optimizing both classification performance and embedding utility.

Dataset:

ITTickets.csv uploaded along with problem statement.

Part 1 : The goal in this part is to prepare the text data. (3 Marks)

- a. Read the text data and lowercasing and EDA
- b. Tokenization, encoding and Text vectorization
- c. Extract features and labels

Part 2 (3 marks):

Build a simple feedforward Neural Network with the below architecture:

1 Embedding layer of shape (,10)

3 Dense layers with 100 neurons and relu activation functions

1 appropriate output dense layer with appropriate activation function

Compile the model and train for 50 epocs (or accuracy score of atleast 75%)

Part 3:

Extract the embedding vector from the model – 1 marks

Visualize the word embeddings in a scatter plot – 1 marks (use PCA)

Predict the class for the below given description (1 marks):

- a. additional monitor hello kindly asking log assign provide monitor please update relevant information after assisting thank kind regards administrator friday pm additional monitor hello please provide additional monitor relocated tower thank regards engineer mob.”

- b. "oracle procurement preparer role needed for louden preparer role needed hi kindly assist kind regards functional analyst thursday pm preparer role needed hi please preparer role please possible kind regards administrator"

Calculate Cosine Similarity of the below pair of words from the extracted word embeddings: (1 marks)

- a. Downtime, Disruption
- b. Configuration, Change
- c. Resolution, Closure

Expected Output :

Jupyter notebook with clear mention of all the members of your team including Name and Bits ID.

There should not be any error cell, and all the cells should be executable.